

4.13 Classwork: Binomial distribution

Paper 1: No calculator

4. [Maximum mark: 5]

In the expansion of $(x + k)^7$, where $k \in \mathbb{R}$, the coefficient of the term in x^5 is 63.

Find the possible values of k .

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Paper 2: Use a calculator

6. [Maximum mark: 5]

Consider the expansion of $(3 + x^2)^{n+1}$, where $n \in \mathbb{Z}^+$.

Given that the coefficient of x^4 is 20 412, find the value of n .

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10 February 2026

4.13 Classwork: Binomial distribution

3. a. A box holds 240 eggs. The probability that an egg is brown is 0.05.

Find the expected number of brown eggs in the box.

[2 marks]

- b. Find the probability that there are 15 brown eggs in the box.

[2 marks]

- c. Find the probability that there are at least 10 brown eggs in the box.

[3 marks]

- 4a. The following table shows the probability distribution of a discrete random variable X .

x	0	2	5	9
$P(X = x)$	0.3	k	$2k$	0.1

Find the value of k .

[3 marks]

- b. Find $E(X)$.

[3 marks]

5. The random variable X has the following probability distribution.

x	1	2	3
$P(X = x)$	s	0.3	q

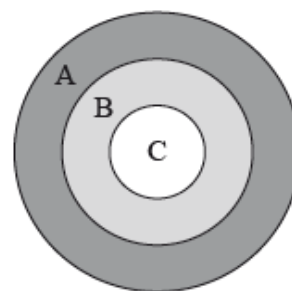
Given that $E(X) = 1.7$, find q .

[6 marks]

- 6a. The following diagram shows a board which is divided into three regions A , B and C .

A game consists of a contestant throwing one dart at the board. The probability of hitting each region is given in the following table.

Region	A	B	C
Probability	$\frac{5}{20}$	$\frac{4}{20}$	$\frac{1}{20}$



Find the probability that the dart does **not** hit the board.

[3 marks]

- b. The contestant scores points as shown in the following table.

Region	A	B	C	Does not hit the board
Points	0	q	10	-3

Given that the game is fair, find the value of q .

[4 marks]

Paper 1: No calculator

2. [Maximum mark: 4]

Consider two consecutive positive integers, n and $n + 1$.

Show that the difference of their squares is equal to the sum of the two integers.

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4. EITHER

attempt to use the binomial expansion of $(x + k)^7$ **(M1)**

$${}^7C_0x^7k^0 + {}^7C_1x^6k^1 + {}^7C_2x^5k^2 + \dots \text{ (or } {}^7C_0k^7x^0 + {}^7C_1k^5x^1 + {}^7C_2k^5x^2 + \dots)$$

identifying the correct term ${}^7C_2x^5k^2$ (or ${}^7C_5k^2x^5$) **(A1)**

OR

attempt to use the general term ${}^7C_r x^r k^{7-r}$ (or ${}^7C_r k^r x^{7-r}$) **(M1)**

$r = 2$ (or $r = 5$) **(A1)**

THEN

$${}^7C_2 = 21 \text{ (or } {}^7C_5 = 21 \text{) (seen anywhere)} \quad \text{ **(A1)** }$$

$$21x^5k^2 = 63x^5 \text{ (} 21k^2 = 63, k^2 = 3 \text{)} \quad \text{ **A1** }$$

$$k = \pm\sqrt{3} \quad \text{ **A1** }$$

Note: If working shown, award **M1A1A1A1A0** for $k = \sqrt{3}$.

[5 marks]

6. METHOD 1

product of a binomial coefficient, a power of 3 (and a power of x^2) seen **(M1)**

evidence of correct term chosen **(A1)**

$${}^{n+1}C_2 \times 3^{n+1-2} \times (x^2)^2 \left(= \frac{n(n+1)}{2} \times 3^{n-1} \times x^4 \right) \text{ OR } n-r=1$$

equating their coefficient to 20412 or their term to $20412x^4$ **(M1)**

EITHER

$${}^{n+1}C_2 \times 3^{n-1} = 20412 \quad \textbf{(A1)}$$

OR

$${}^{r+2}C_r \times 3^r = 20412 \Rightarrow r = 6 \quad \textbf{(A1)}$$

THEN

$$n = 7 \quad \textbf{A1}$$

METHOD 2

$$3^{n+1} \left(1 + \frac{x^2}{3} \right)^{n+1}$$

product of a binomial coefficient, and a power of $\frac{x^2}{3}$ OR $\frac{1}{3}$ seen **(M1)**

evidence of correct term chosen **(A1)**

$$3^{n+1} \times \frac{n(n+1)}{2!} \times \left(\frac{x^2}{3} \right)^2 \left(= \frac{3^{n-1}}{2} n(n+1)x^4 \right)$$

equating their coefficient to 20412 or their term to $20412x^4$ **(M1)**

$$3^{n-1} \times \frac{n(n+1)}{2} = 20412 \quad \textbf{(A1)}$$

$$n = 7 \quad \textbf{A1}$$

Total [5 marks]